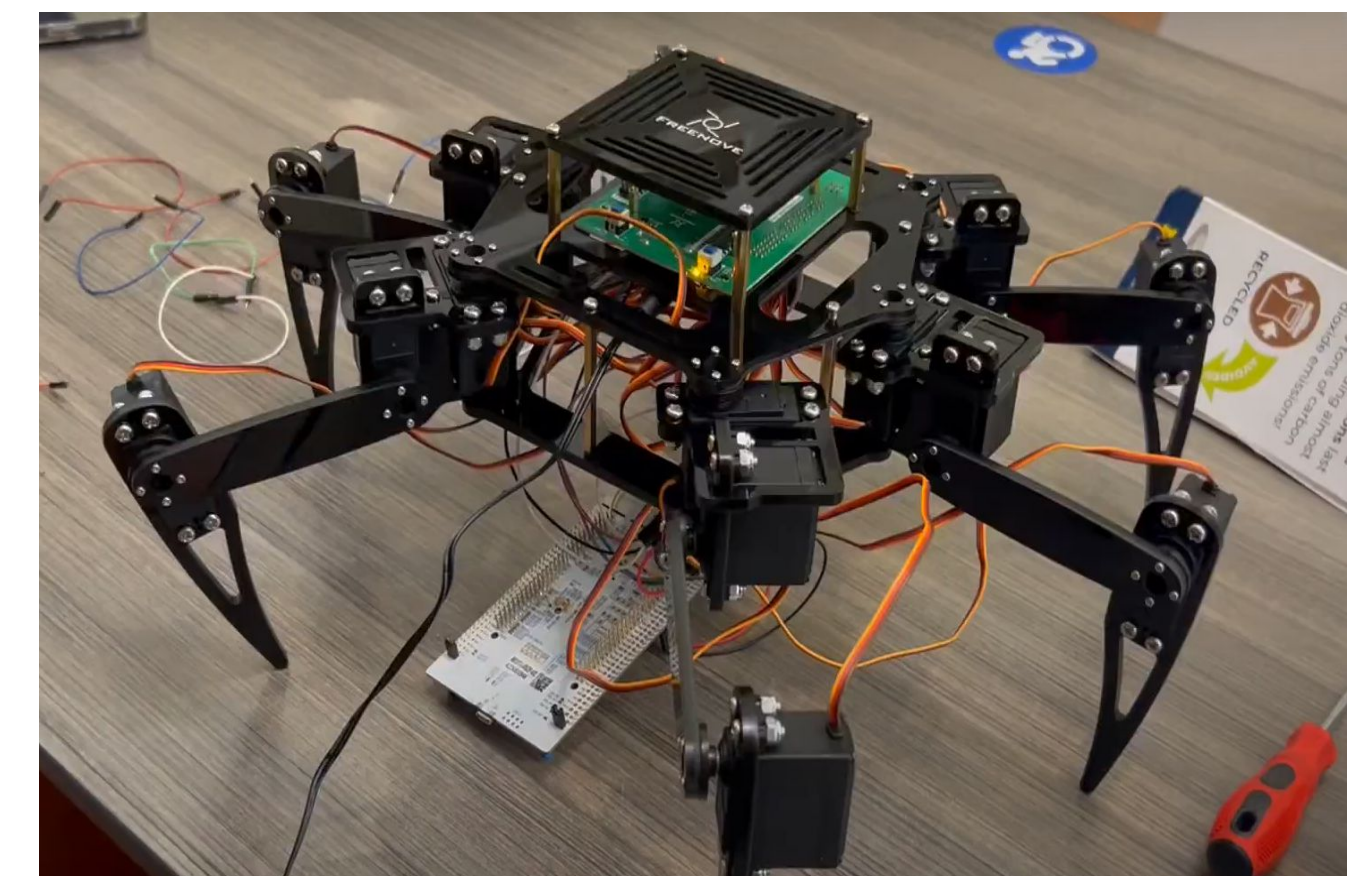


Spider-People

Richard Feng, Anthony Tan, Lena Trang
 {rrfeng, anttan, ltrang}@umich.edu

Introduction: A Novel RC Vehicle to Simulate “All Walks of Life”

- Many remote controlled vehicles are cars with limited applicable environments and unsuited to rough terrain
- Exploring various modes of movement advances the understanding of the capabilities and limitations of robot navigation in difficult terrains, e.g. mountainous environments
- Navigation through difficult environments can be used for retrieval and delivery
- Robots with novel mobility schemes are visually interesting and can inspire kids to learn more about robotics



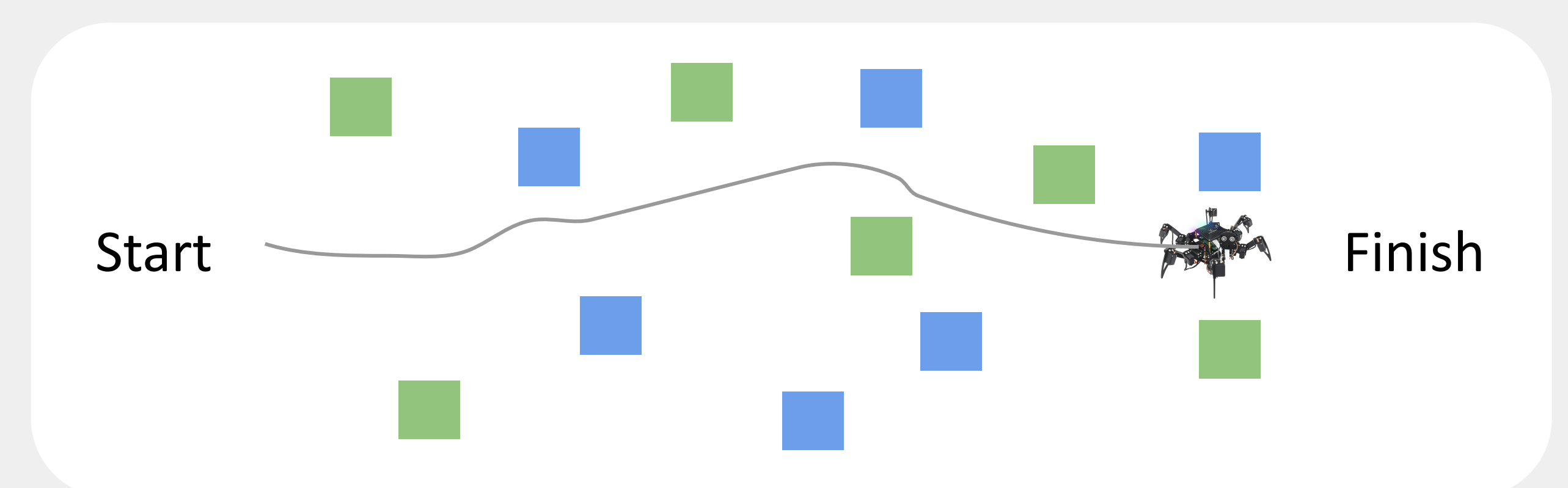
Problem Statement: Walking Without Wheels

Goal:

- Create a remote controlled robotic platform, **Spider**, with three walking modes and shows users when in unstable positions
- Showcase the walking gaits with an obstacle course

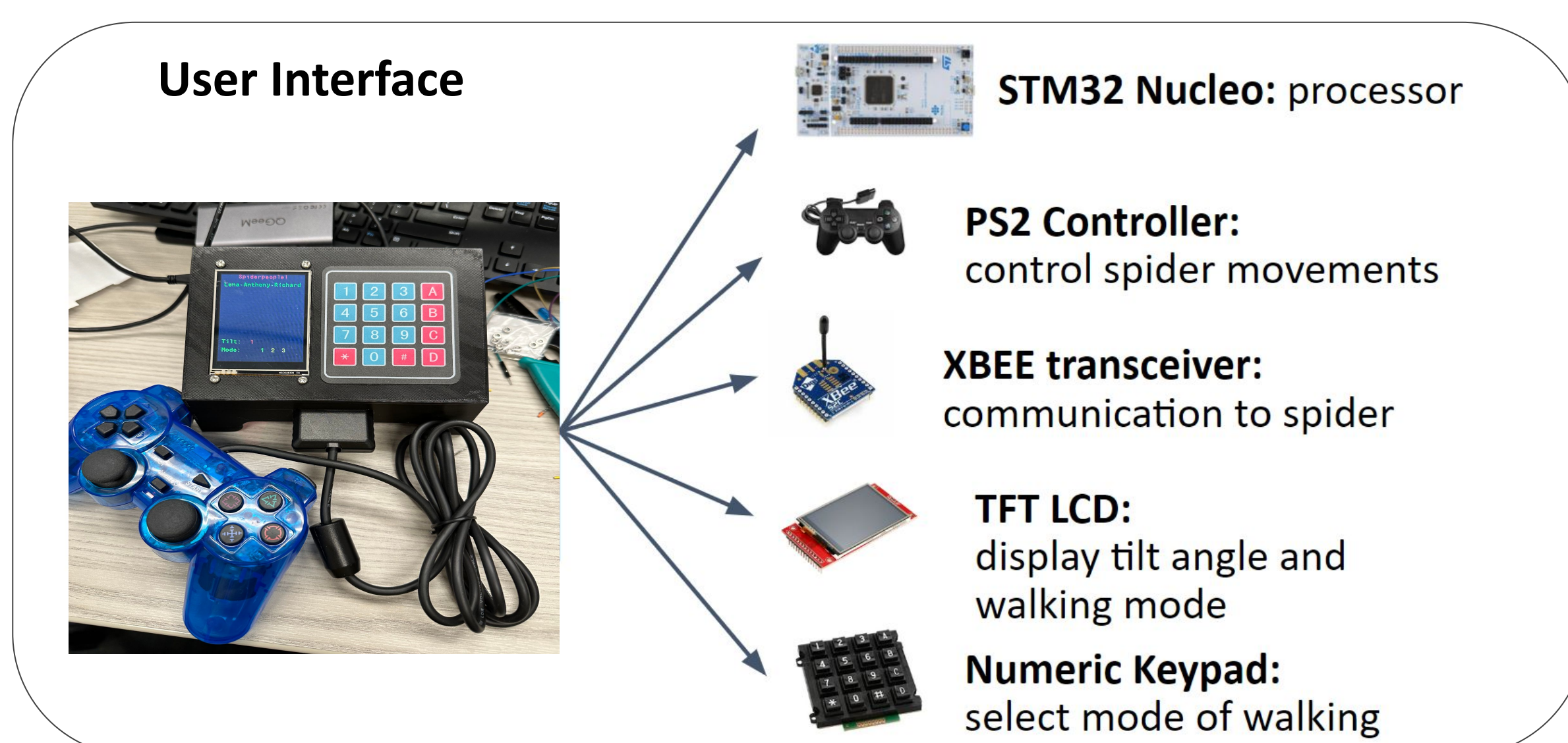
Design:

- Handheld controller to control Spider’s movement
- Keypad to change Spider’s walking gait
- LCD Display to show Spider’s walking mode and tilt of body

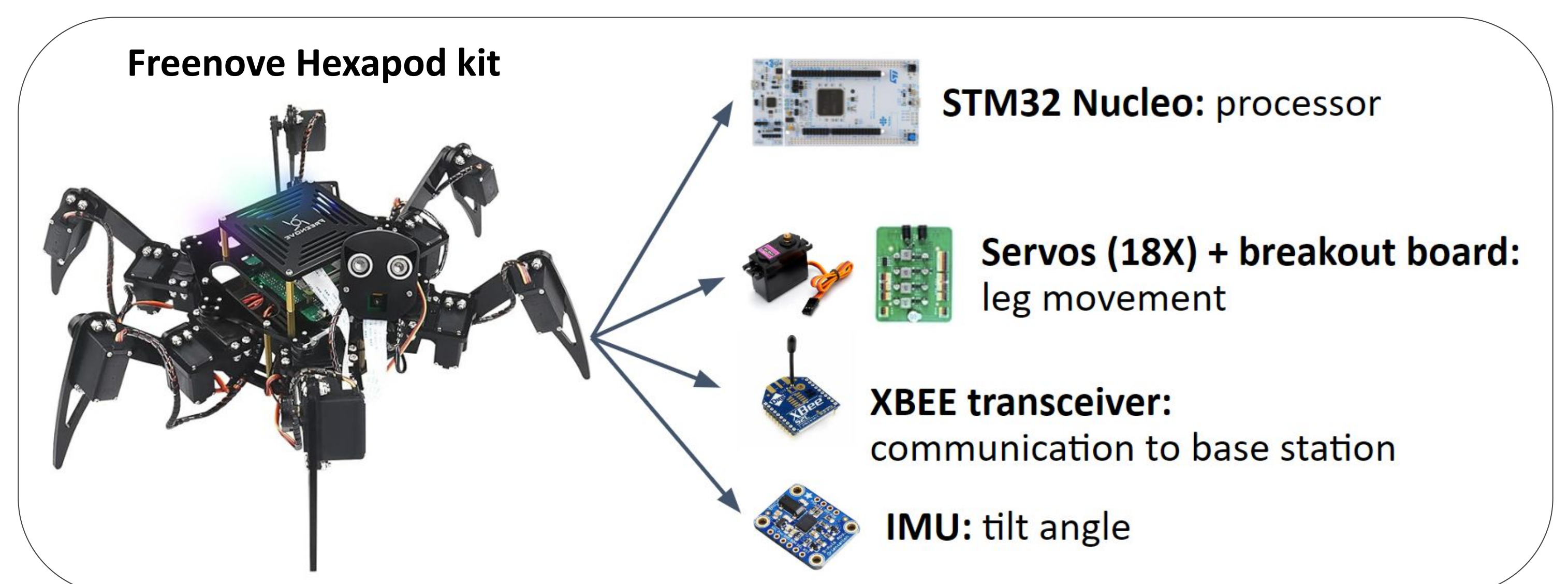


Solution: Remote Control Robotic Spider

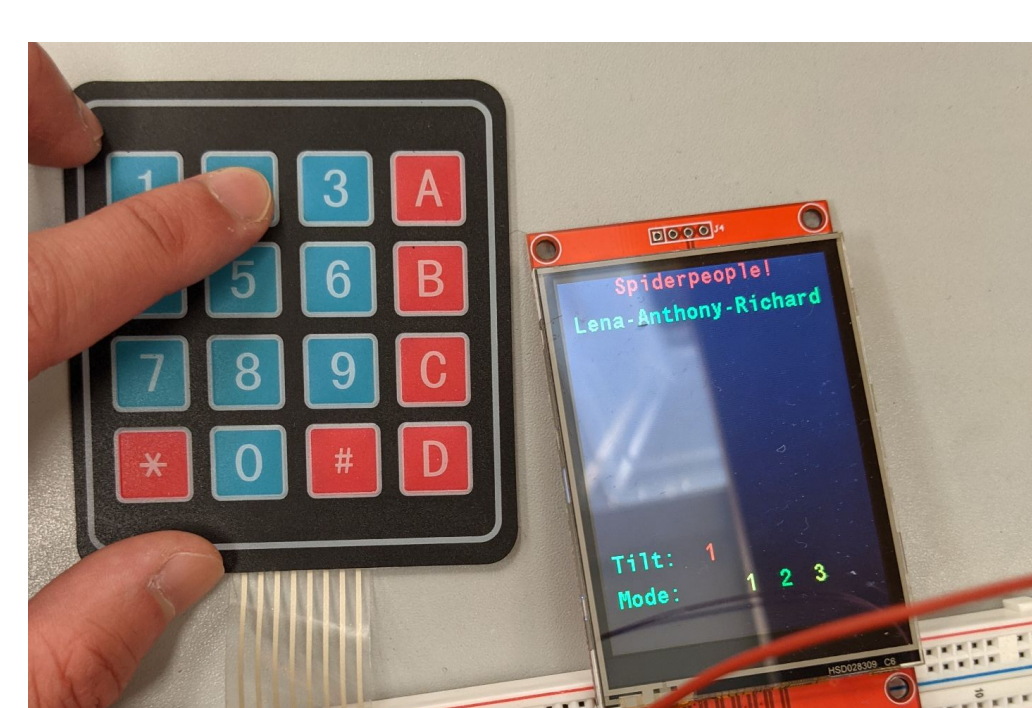
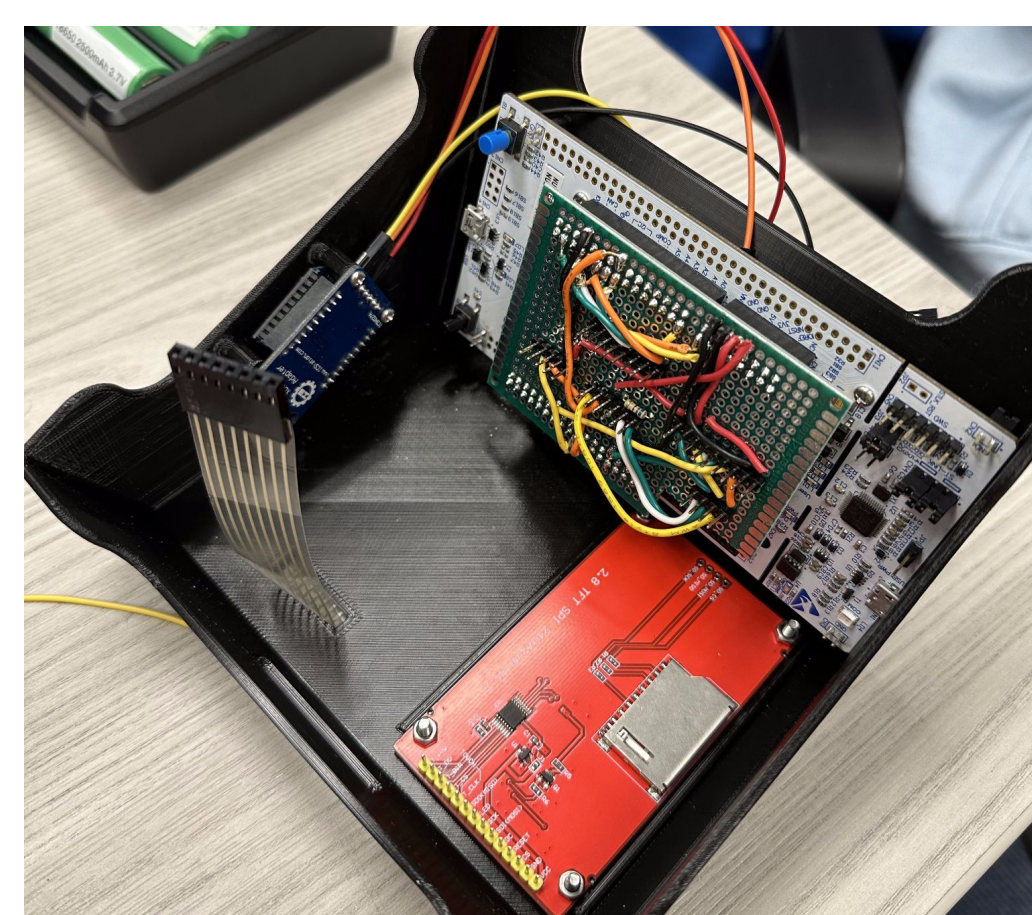
Base Station



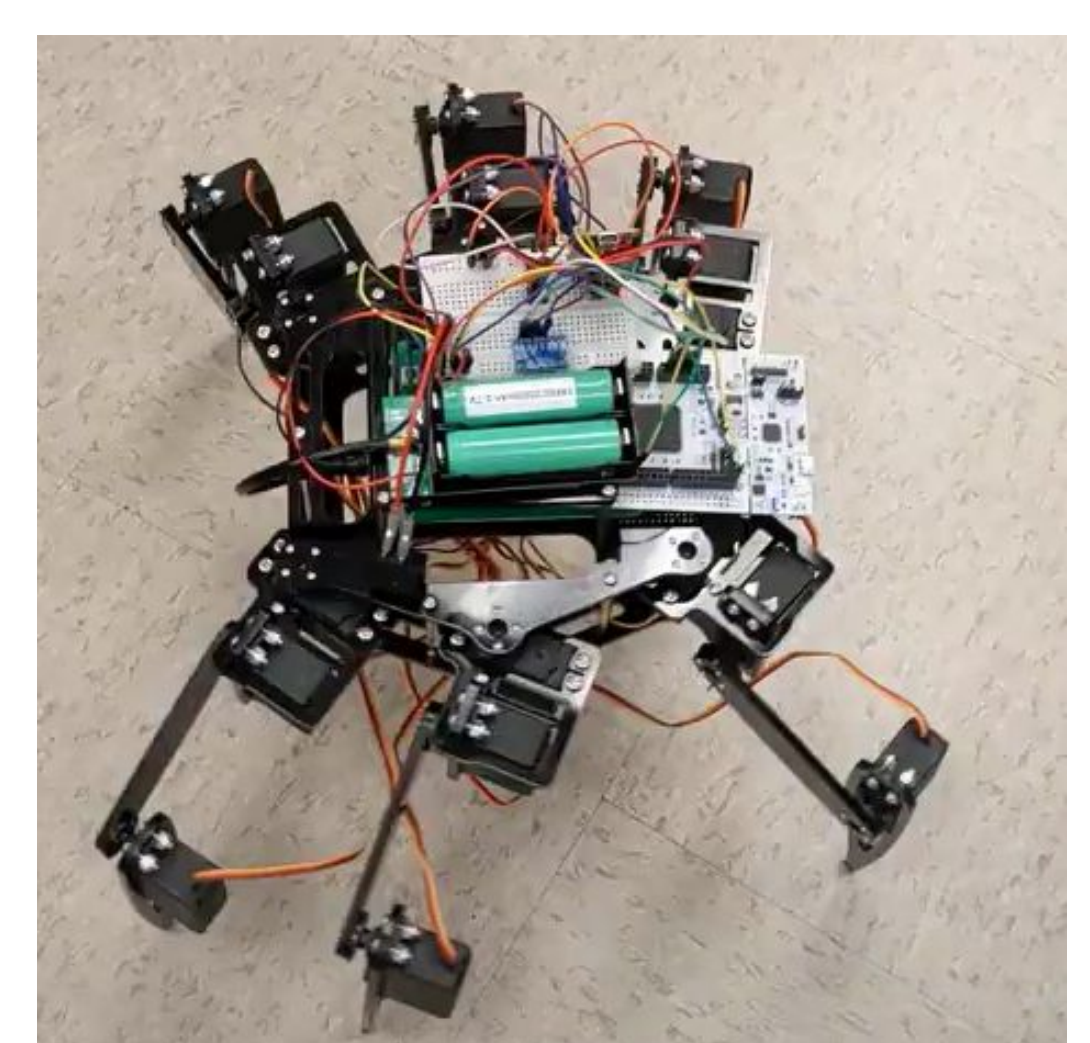
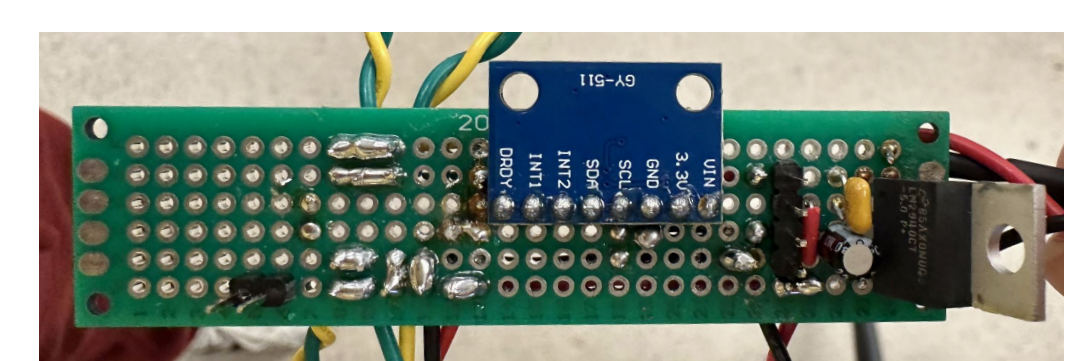
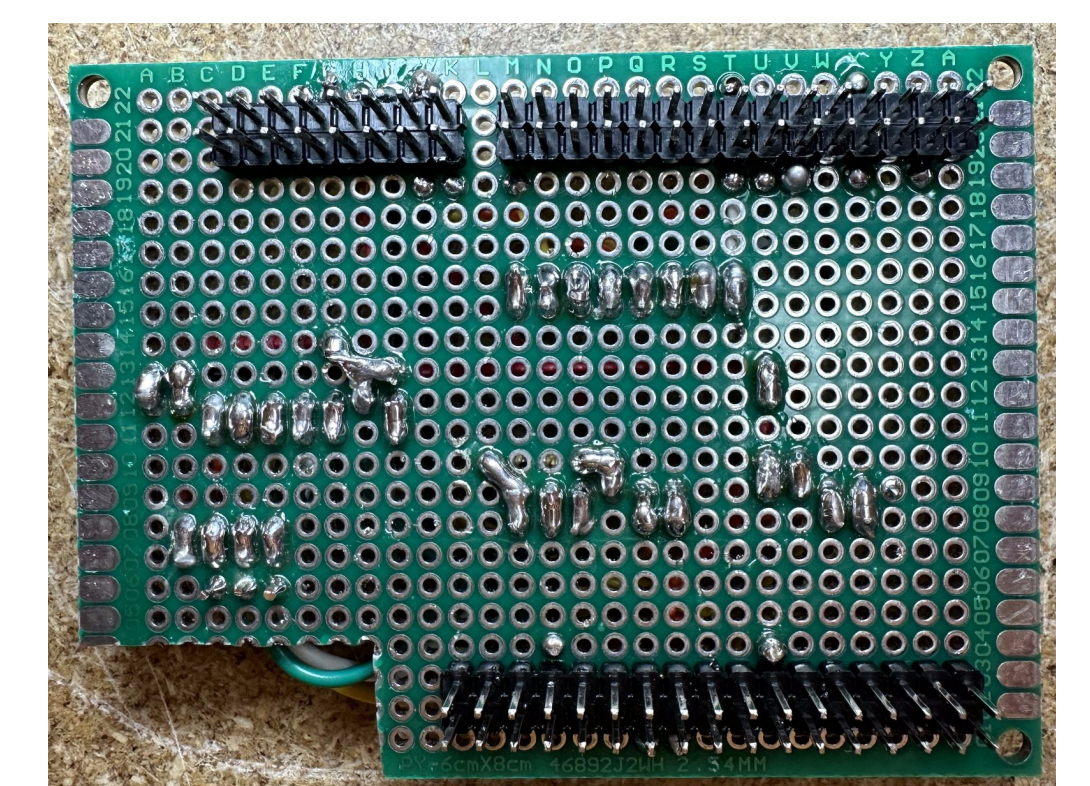
Spider



- Walk the Spider with **controller analog sticks**
 - **Left** for forward/back
 - **Right** for turning
- A pull-up resistor and **SPI interface** were used to read controller data
- **3D-printed housing** stores all electronic components on the base station
- Use the **keypad** to change walking modes
- **LCD screen** uses an **SPI interface** to display Spider’s tilt and selected walking mode
- A technical challenge was receiving controller and keypad data which we solved with **interrupts** as polling both was too slow



- **Soldered prototyping boards** designed as nucleo shields protect the base station and Spider
- **XBEE transceivers** send data serially from the base station to the Spider and vice versa
- An onboard **IMU** produces the angle of the body, which is displayed to give a sense of the Spider’s stability
- The Spider moves its legs via **servos** and an **I2C interface**
- **Hardware timers** enable precise control of servo movements
- A technical challenge was programming the servos to **simulate** multiple **walking styles**



Ready to Try Walking like a Spider?